

Chapter 2

Landslide causes and triggers

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2.1 Introduction

On the 8th of August 1979 in Abbotsford, New Zealand, several weeks of preliminary movements culminated in the rapid failure of an approximately five million m³ block slide in a residential area. The cost of the Abbotsford Landslide amounted to some NZ\$ 10–13 million and included the loss of 69 houses, but no human casualties (Hancox, 2008). As with the Abbotsford Landslide case, when any landslide or other hazard results in financial or human losses, an element of human blame will often arise, particularly in cases where insurance companies need to make loss adjustments. This, and the desire to prevent similar disasters in the future, necessitates investigation into the causes of a landslide. Furthermore, to gain an appreciation of the role of landslides in landscape systems and the geomorphic evolution of landscapes, it is necessary to have some knowledge of the factors that determine where and over what timescales different types of landslides are likely to occur or have occurred. As with almost every other landslide, the cause of the Abbotsford Landslide cannot be put down to a single factor; its failure was the result of the convergence of a series of unfavourable conditions, both natural and human induced. Hancox (2008) reports on the results of a Government Commission of Inquiry which set out to identify the causes of this landslide. The following factors were considered to have contributed:

1. Unfavourable geology, consisting of weak clay layers in gently tilted strata.
2. Removal of toe support over the preceding millennia by natural fluvial erosion and later by artificial quarrying 10 years prior to failure.
3. A long-term rise in groundwater resulting from a leaking water main above the slide area, clearance of vegetation and a natural increase in rainfall over the preceding decade.
4. Urban development potentially adding a load surcharge.

5. Rainfall during the last few days of movement, which by itself may not have been the difference between failure and stability, but likely influenced the timing and nature of the landslide's final rapid movement.

Despite all of these causes, the disaster, rather than the landslide event, stems from the decision to build (perhaps unknowingly) on terrain susceptible to landsliding without adequate precautionary mitigation measures in place. Although land use decisions were a root cause of the disaster, the purpose of presenting this case study here is to highlight that, as with many landslides, none of the recognised factors alone would have been sufficient or necessary to cause the landslide. Assigning blame to any person, organisation or decision is therefore challenging. Yet, knowledge of the various factors that can cause landslides is invaluable for hazard mitigation. This knowledge helps to identify and avoid human activities that may affect, or be affected by, hillslope instability, and it improves the capability for predicting natural landslides. For example, providing effective advanced warnings for debris flows requires appropriate triggering thresholds to be set. At which level these thresholds are set and indeed the identification of the triggers themselves requires thorough understanding of the various factors causing and triggering debris flows. These factors may include the availability of debris to be mobilised, the geotechnical properties of the debris, antecedent soil moisture conditions, rainfall magnitude and intensity and the condition of vegetative cover. Without sufficient knowledge of contributing instability factors and the appropriate application of that knowledge, hazard mitigation efforts may be wasted.

A great deal of scientific observation and enquiry over the last few decades has contributed to more in-depth classification, rationalisation and understanding of the natural and human causes and triggers of landsliding. This has stemmed largely from the disciplines of engineering geology, soil and rock mechanics and geomorphology, but there is now a well-established field of landslide science. This chapter does not attempt to list or comment on the history of significant developments in landslide research (which has been done by others; e.g. [Crozier, 1986](#); [Petley, 2011](#)), nor does it attempt to describe every known landslide cause. Instead, the work focuses on causes that are of high importance and remain fruitful objectives of scientific research. To begin, the concept of hillslope stability is introduced to provide a conceptual basis for understanding the effects of various hillslope conditions and destabilising processes. This is followed by a brief overview of the main types of landslide causes and triggers and the current understanding of these. Finally, a brief synopsis of landslide causes and opportunities for further research is presented.

2.2 Concept of instability

Destabilising stresses are present within all slopes. Whether or not these stresses (driving stresses) are capable of triggering failure of a given hillslope at a given moment in time will depend on the relative magnitude of stresses that resist the tendency for failure; these opposing stresses can be referred to as resisting stresses. Both driving and resisting stresses can change over time. Thus, ‘stability’ is both a relative term and one that holds meaning over a certain time period. The vast majority of existing slopes are stable at this moment in time (otherwise they would be in the process of failing), but every hillslope has the potential to fail at some future time. The magnitudes of driving and resisting stresses are the result of stability factors, which can be defined as any phenomena that control or influence the forces that determine the stability (Crozier, 1986). Some stability factors are inherent to the hillslope and unchanging (e.g. lithology), whilst others may be transient and their influence may come and go or vary in magnitude (e.g. porewater pressures).

Stability can be quantitatively assessed by measuring the balance of driving and resisting stresses. The development of shear stresses drives a tendency for failure for most landslide types (with the exception of toppling). The resisting stresses result from reactionary stresses, and can be considered as the mobilised shear strength of the hillslope with respect to the shear stresses. Mobilised strength refers to only the strength that is used to resist movement and is distinguished from the ‘total’ strength of the hillslope, which is arguably an impossible quantity to assess. For example, the movement of an intact sliding block (e.g. a rockslide) is influenced by the strength generated between the sliding surfaces, not the internal strength of the block itself; therefore, only the strength of the failure surface needs to be known to assess the stability of the sliding block. However, the assumption that landslides involving sliding are governed by shear stresses alone is an oversimplification. Other (e.g., compressional and tensional) stresses most likely also play a role, especially at the boundaries of the sliding mass and where the landslide mass moves over irregular surfaces.

The balance of strength and driving forces is often expressed as a ratio, commonly referred to as the factor of safety (FoS):

$$\text{Factor of safety (FoS)} = \frac{\text{resisting stresses (or mobilised shear strength)}}{\text{driving stresses (or shear stresses)}}$$

The FoS equation can be populated with a wide range of parameters which influence the resisting and/or driving stresses. The choice and determination of these parameters depends on the problem being addressed, the physical conditions and processes expected for the hillslope and the degree of simplicity or

complexity being achieved. The method of assessing stability by determining the FoS is referred to as limit equilibrium analysis. Examples and explanations of this process and the selection of relevant parameters and variables are provided in numerous texts on hillslope stability (e.g. Duncan, 1996; Norrish and Wyllie, 1996; Selby, 1993).

The effect of gravity is of fundamental importance in the stability of a hillslope: The strengths and stresses operating in a slope depend on the way that gravity causes masses within a slope to interact. The weight of material can be resolved into stresses acting normal and parallel to contact surfaces (Fig. 2.1). The greater the normal stresses, the greater the frictional shear strength of the potential failure surface. The relationship between shear strength and the normal stresses is often considered to be linear and governed by the Mohr–Coulomb failure criterion. The greater the stresses acting parallel to the potential failure surfaces, the larger the shear stress. Many of the stability factors discussed below influence the stability of a hillslope by directly affecting or altering, acutely or chronically, the shear stresses and/or shear strength. Thus, it is useful to keep the relationship between these two stresses in mind.

The term ‘slope stability’ can have different meanings depending on the context of its use or the objectives of its user. It often refers to the inherent stability or FoS, as determined by the static physical hillslope properties; for example, all other stability factors being equal, a high embankment is less

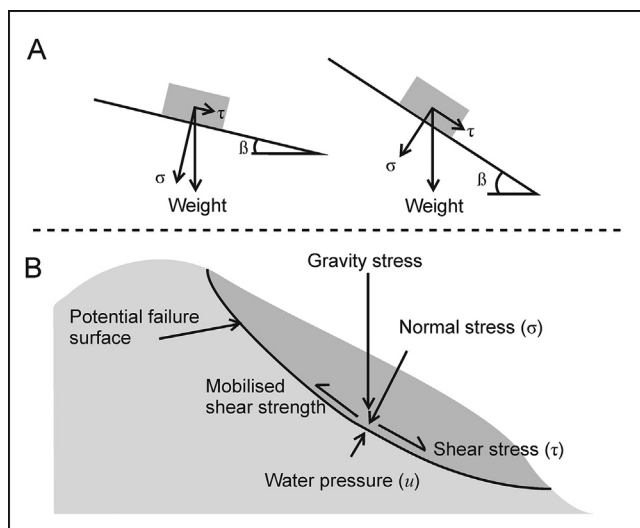


FIGURE 2.1 (A) The influence of slope angle (β) on the relative magnitudes of shear (τ) and normal stress (σ). (B) Stresses acting along a potential failure surface. Adapted from Selby (1993), by permission of Oxford University Press.

stable (i.e. lower FoS) than a low embankment. Landslide susceptibility, which is a measure of the inherent stability of a hillslope or distribution of hillslopes, treats the term stability in this way, without any consideration of the likelihood of failure. Alternatively, *slope stability* can, sometimes more usefully, be a measure of the probability of a failure of an individual hillslope, which requires consideration of both the stability factors (i.e. slope stability in the previous sense), and the likelihood of a critical failure threshold being exceeded in a given time span.¹ Assessing the stability of the above two embankments in this sense requires knowledge of the potential failure triggers and their likelihood during the time period of interest for each embankment. That knowledge must include the probability that a trigger of sufficient (critical) magnitude to induce failure will occur. It might be worth noting that the critical magnitude required is itself not static in time, but fluctuating over some range, so its determination may also warrant a probabilistic approach. The higher embankment is thus not necessarily more likely to fail than the lower embankment, because it may exist in an environment less likely to produce a trigger capable of exceeding the critical threshold for stability for that slope.

In considering the causes of mass movements, a distinction must also be made between the terms ‘failure’ and ‘movement’. Some mass movements only ever exist as a discrete event with the failure resulting in the first and only movement of that material; for example, a rockfall. Other mass movements can exist in a state of transient stability for a long time after failure is first initiated and be affected by entirely different movement triggers or controlling factors to those that initiated the first failure. These slopes may experience discrete periods of instability (movement) and periods of stability (no movement), as is often the case for slopes affected by large, slow-moving rockslides. Other mass movements may evolve from an initially very slow creeping mass and begin to accelerate as the failure surface develops, culminating in a catastrophic final failure; such mass movements are termed progressive failures (Petley et al., 2005). In such cases, it is not clear to which part of this process the term ‘failure’ applies; the final catastrophic failure only, the initiation of creep or the entire process (which may take place over an indeterminate amount of time). Perhaps it depends on the scale at which a hillslope is observed. Petley (2011) explains a distinction between a local and a global FoS for a hillslope. The global FoS applies to an entire landslide body, although this in itself may be difficult to define for landslides involving movement of multiple (nested) blocks. Failure of the entire mass occurs if the

1. The parameters used in factor of safety equations can also be treated probabilistically, in recognition of the difficulty and uncertainty of measuring the absolute value of the parameters (e.g. the friction angle value), with the chosen value, at best, representing a range of possible values that may exist for any one slope (e.g. El-Ramly et al. 2002). Additionally, even the type of failure mechanism may be assessed probabilistically (e.g. Hack et al. 2002).

global FoS falls below unity. However, even if the landslide as an entire mass may be stable (global FoS > 1), there may be parts of the landslide body that are locally unstable. Petley points out that those unstable parts of the landslide mass will locally undergo deformation, and cumulative deformation of these unstable parts may lead to progressively greater reductions in the global FoS (see Section 3.2).

Terms such as ‘stable’, ‘marginally stable’ and ‘actively unstable’ have been introduced in recognition of the need to classify the stability state of landslides as a function of the likelihood of failure (Fig. 2.2) (Glade and Crozier, 2005). In this context, ‘stable’ refers to a hillslope that exists in an environment which is unlikely to produce a trigger sufficient to cause failure or movement of that slope (Fig. 2.3). A ‘marginally stable’ hillslope is one for which a trigger of sufficient magnitude to cause failure can be expected to occur in the prevailing environmental conditions (Fig. 2.3). Actively unstable hillslopes are those that are already unstable and actively moving. A hillslope may shift between these states with a change in any of a multitude of stability factors. Any factor (or collection of factors) that contributes to the reduction in stability to a marginally stable state can be considered to be a landslide cause (or causes). If any of these factors changes the hillslope to an actively unstable state, it is considered a trigger. Whilst there can be multiple causes, there is usually only one trigger (Varnes, 1978) (Fig. 2.3). Others (e.g. Varnes, 1978 and references therein) have previously commented that a landslide trigger is often only the final action that initiates a marginally stable hillslope, and it may be quite trivial in magnitude (Fig. 2.3). This can be likened to the idiom ‘the straw that broke the camel’s back’. An astonishing example of movement

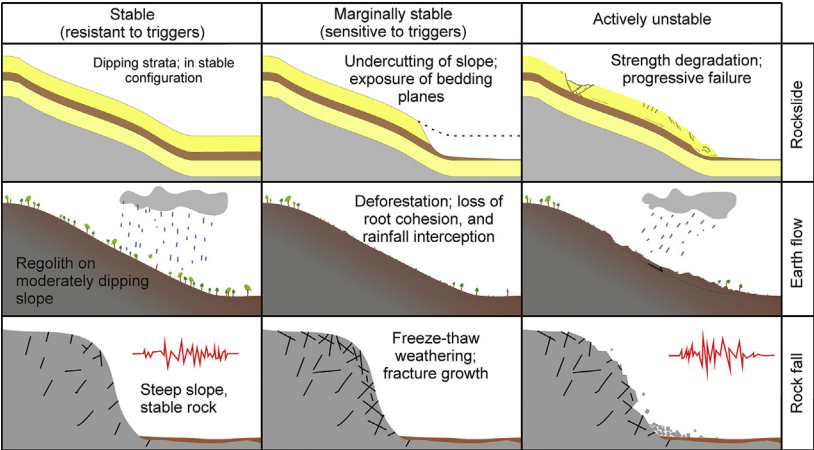


FIGURE 2.2 The relationships between stable, marginally stable and actively unstable slopes and the interactions of stability factors for a range of landslide types. Based on Glade and Crozier (2005).

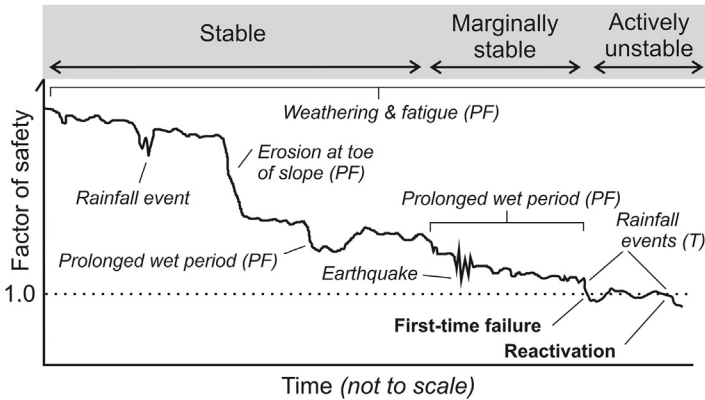


FIGURE 2.3 Schematic representation of the reduction in the FoS of a hillslope, both through gradual (e.g. weathering) and more discrete (e.g. undercutting) preparatory factors (PF). The resistance of the hillslope to external processes (potential triggers) lowers through time, eventually bringing the slope to a marginally stable state, whereby it is sensitive to triggering (T) by such events. First-time failure may result in complete failure of the unstable mass (e.g. a rockfall), returning the hillslope to a more stable state. Or, as shown here, the hillslope (e.g. as for a long-lived earth flow) may exist in an actively unstable state, hovering about equilibrium. In this case, it remains sensitive to sustaining factors (e.g. groundwater) or can reactivate from time to time. Adapted from *WG/WLI (1994)*.

being triggered by a seemingly trivial force is the Slumgullion landslide in the United States of America (*Schulz et al., 2009*); when the landslide is critically stable due to a combination of other unfavourable stability factors, a tidal-induced state of low atmospheric pressure is sufficient to reduce shear strength and promote sliding.

The stability factors can be divided into preconditioning, preparatory and triggering factors (*Glade and Crozier, 2005*) (*Fig. 2.2*). Preconditioning factors are those that influence the inherent strength of the hillslope and are generally considered to be temporally unchanging (over human timescales). Factors that reduce the stability over time but do not cause failure or movement are termed preparatory factors, and factors that change a hillslope to an actively unstable state (i.e. initiate failure or movement) are termed triggers. Descriptions and examples of preconditioning, preparatory and triggering factors are provided in the following section and examples of each are presented in *Table 2.1*. Several of these processes can act as either a preparatory factor or a trigger, or both, depending on their degree of activity and the margin of stability of a hillslope (*WG/WLI, 1994*). For example, even if an earthquake or rainstorm does not trigger a failure, the event may be severe enough to cause deterioration in the strength of slope material, thus reducing the stability of the hillslope and potentially allowing a subsequent earthquake or storm of a similar or smaller size to trigger failure (e.g. *Parker et al., 2015*; *Petley, 2011*).

TABLE 2.1 Examples of preconditioning, preparatory or triggering factors and examples of the processes involved. Process examples have been grouped into geomorphological, physical and human processes following the classification of WG/WLI (1994) . More comprehensive lists of landslide causes and triggers can be found in Varnes (1978) , Cruden and Varnes (1996) and WG/WLI (1994) .			
Preconditions			
Sloping ground (or elevated ground in the case of lateral spreading of flat ground)			
Plastic weak material			
Sensitive material			
Collapsible material			
Weathered material			
Sheared material			
Jointed or fissured material			
Adversely oriented mass discontinuities (including bedding, foliation, cleavage, faults, unconformities, flexural shears and sedimentary contacts)			
Contrast in permeability and its effects on groundwater			
Contrast in stiffness (stiff, dense materials overlaying weak plastic materials)			
	Processes		
Preparatory factors	Geomorphological	Physical	Human
Increase in slope height or steepness	Growth or tilting from tectonism, volcanism or glacial rebound. Hillslope erosion		Mining Slope profiling
Debuttressing	Fluvial, marine or glacial erosion/ undercutting glacier retreat		Unloading of toe of slope; excavation, road cuts
Exposure of potential failure surface	Fluvial, marine or glacial erosion/ undercutting		Slope excavation
Reduction in inherent strength	Soil piping or karstification	Weathering stress-induced fatigue	Tunnelling, underground mining and deforestation
Loading of the slope	Gradual build-up of sediments		Construction or emplacement of engineering fill or waste deposits

Continued

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	Processes		
Preparatory factors	Geomorphological	Physical	Human
Long-term increase in groundwater levels		Natural climate change	Infiltration from storm water or broken pipes Irrigation Removal of vegetation Anthropogenic climate change
Triggers	Geomorphological	Physical	Human
Rapid increase in porewater pressures		Rainfall Thaw of snow/ice	Undrained loading from rapid emplacement of fill or heavy loads Artificial reservoir filling
Drawdown of groundwater	Natural dam breach	River flood recession	Rapid lowering of reservoir level
Transitory applied stresses	Passage of rock avalanche	Earthquake wind	Machinery vibrations
Reduction in strength		Permafrost degradation Weathering Stress-induced fatigue	
Loading of the slope	Other landslides	Vegetation Precipitation	Construction materials Snow sports (snow loading)

Therefore, earthquake and rainfall events can act as both preparatory factors and triggers. Some preconditioning and preparatory factors may also be similar. For example, nonzero slope is a precondition for all landslides, but a change in hillslope angle is considered a preparatory factor. The landslide causes and triggers listed in [Table 2.1](#) include conditions or processes acting

both internally and externally to the hillslope and can reduce the FoS either through an increase in the shear stresses, a reduction in the shear strength or in some cases both (Terzaghi, 1950; Varnes, 1978; WG/WLI, 1994).

2.3 Stability factors

2.3.1 Material strength and topography

The strength of material preconditions the stability of all slopes. Material strength determines both the height and angle that a hillslope can sustain without failure. The influence of material strength and geometry on hillslope stability can be inferred from spatial patterns of landslide distributions (e.g. Dai et al., 2011; Keefer, 2000; Roering et al., 2005) (Fig. 2.4). These relationships allow, to a limited degree, determination of landslide susceptibility. Material type (e.g. soils, soft rock or brittle rock) and hillslope geometry also influence the type of landslide that may be expected (e.g. soil flow, rotational slide and rock avalanche), and knowledge of these parameters can also be used to help predict the type of landslides expected (Cruden and Martin, 2013; Gerrard, 1994). Thus, knowledge of material type, hillslope geometry and the orientation of structural features relative to slope aspect is essential for assessing slope stability.

Material strength is derived from the strength of the particles or crystals, the inter-particle contact forces, the structure produced by fabric or discontinuities and apparent cohesion provided by vegetation, moisture or engineering materials. Materials are traditionally divided into (indurated) rock and soils, the latter of which can be further sub-divided into cohesive and noncohesive soils. Mohr–Coulomb strength parameters (friction and cohesion) are routinely assessed for each type of soil or rock present at a site; for the latter, an equivalent Mohr–Coulomb failure criterion that incorporates the effects of discontinuities (e.g. joints) should be used (e.g. Hoek et al., 2002). In rock masses, discontinuity mapping usually involves quantifying density, orientation and strength and grouping into sets where they are systematic (i.e. all sharing similar characteristics). In soils, it may be more appropriate to quantify the depth, orientation, strength and permeability of distinct compositional units.

Discontinuities are perhaps the most important determinant of strength, particularly when dipping out of the hillslope. They arise from compositional changes (namely bedding or weathering horizons) or structural weaknesses such as cleavage, foliation, fractures and faults. There are two main effects that discontinuities and compositional changes have on the stability of a hillslope:

1. They provide structural weaknesses below the strength of the intact material, which may become preferential failure surfaces, particularly when planar, or influence failure mechanisms (Stead and Wolter, 2015). For example, the soft-rock Neogene landscapes of New Zealand host many

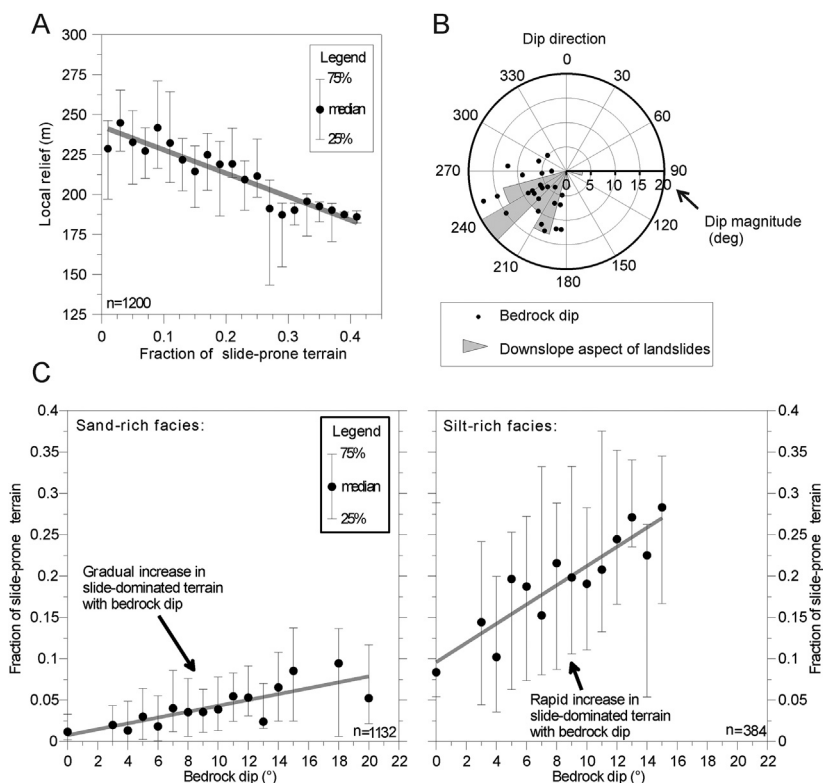


FIGURE 2.4 Example from part of the Oregon Coast Range, U.S.A. showing the relationships between geological and geomorphological conditions and landslide distribution. (A) Relationship between relief and landslide abundance in terrain prone to deep-seated landsliding; (B) Strong correspondence between bedding orientation and the failure of landslides; (C) Moderately strong relationship between landslide abundance and the angle of bedding for sand-rich terrain prone to deep-seated landsliding (left) and a stronger relationship for weaker, silt-rich sedimentary rocks (right). Reproduced from [Roering et al. \(2005\)](#) with permission by the Geological Society of America.

thousands of large landslides; there, it is the presence of clay layers, with very low friction coefficients, within the Neogene sandstones and mudstones that result in landsliding on gently tilted and folded dip slopes ([Thomson, 1982](#); [Williams et al., 2021](#)). However, where bedding is horizontal or dipping into the slope, the rock forms steep cliffs, as a result of a relatively high cohesive strength, with a sparsity of tectonic joints.

2. Discontinuities or compositional changes can allow pathways for water seepage or create permeability boundaries, both of which can influence the porewater conditions of the material and modify the hillslope stresses (Section 3.3.1).

A clear relationship between material strength, hillslope geometry and landslide susceptibility may not always exist. For example, one of the largest landslides to develop in the greywacke sandstone of the Central Southern Alps of New Zealand occurs in a location with relatively high rock mass strength. Here, the 150–200 M m³ Mueller Rockslide is inferred to be structurally controlled, having taken advantage of relatively planar bedding surfaces on the limb of a large anticline (Cody et al., 2020). Elsewhere in the greywacke terrain, the folding and jointing is possibly too intense to permit extensive failure surfaces to develop, and consequently the dominant mode of failure is rockfall governed by the rock mass strength rather than structural controls. Further, in an analysis of earthquake-induced landslide distributions arising from the 1989 Loma Prieta earthquake in California, Keefer (2000) demonstrated a complex relationship between material type and failure densities. Although the most indurated rocks generally had a lower propensity of landsliding than less indurated rocks, the least indurated rocks did not have the greatest number of landslides. As well as this, when ranked by estimated or measured shear strength parameters (rather than rock type), there was no statistically significant relationship with landslide density. Although part of the reason for the latter may be problems with the sampling resolution (Keefer, 2000) or variable unaccounted site effects (Section 3.4), it may also be partly explained by the concept of strength equilibrium or threshold hillslopes (*sensu* Selby, 1982). If so, an increase in hillslope stresses (e.g. earthquake shaking) may be equally likely to trigger failure of any of these slopes.

Strength equilibrium hillslopes are slopes in which the gradient is in equilibrium with the present stress conditions and strength of that hillslope. This concept implies that stronger rock masses can attain and maintain steeper slope gradients than weaker rocks, and thus the steepest slopes in a landscape are often composed of the strongest rock (and vice versa) (e.g. Korup and Weidinger, 2011) and are not necessarily the most unstable. Slopes that have undergone erosional steepening are often said to be ‘over-steepened’ (with respect to their strength equilibrium). Over-steepening of strength equilibrium hillslopes, through natural erosion or engineering works, makes the hillslopes unstable or increases their sensitivity to triggers. In a classic study by Savigear (1952) using space-for-time substitution, it was shown that once a disturbing force (in this case coastal erosion) has been removed, over-steepened hillslopes adjusted over time to a more gentle gradient more in line with their strength equilibrium; but noting that as the slopes lose strength from weathering, the threshold hillslope angles are also declining. A well-recognised process of natural hillslope steepening is by glacial erosion of valley sides; glacial over-steepening is a commonly attributed preparatory factor for instability of previously glaciated slopes. However, whether or not a hillslope is over-steepened depends on whether its strength equilibrium has been exceeded, and Augustinus (1995) argued that this needs to be quantified to be able to make such a determination. By quantifying the rock mass strength of

glaciated slopes in Fiordland, New Zealand, [Augustinus \(1995\)](#) found that most slopes in Fiordland today are in strength equilibrium, and despite the history of glacial erosion leaving behind very high hillslope gradients, most slopes are not over-steepened. Hillslopes that were over-steepened were likely to have already adjusted back to strength equilibrium. In general, hillslopes that are in or close to strength equilibrium might be expected to respond quite quickly to any increase in slope angle (from e.g. glacial erosion or fluvial incision), by adjusting their form. [Larsen and Montgomery \(2012\)](#) demonstrate that in high relief parts of the Himalaya, for example, threshold slopes respond to increases in uplift and river incision with an increase in the rate of hillslope failure activity, rather than by increasing slope angle.

Independently of each other, hillslope angle and material strength are not reliable predictors of stability; but, if both material strength (including the strength and orientation of discontinuities) and slope angle can be quantified and assessed together, then susceptibility determination can be powerful. These are the two basic parameters of all hillslope stability models and regional landslide susceptibility maps, whether implicit (i.e. measured) or explicit (i.e. evaluated statistically). Recent advances in remote sensing (e.g. satellite, airborne and terrestrial distance ranging) have permitted very high-resolution topographic models of slopes at various scales, which allow accurate measurements of hillslope gradient. Material strength, on the other hand, is more difficult to directly measure. Therefore, it is seldom assessed at a resolution sufficiently detailed to detect strength changes within geological units at a regional scale. Instead, most susceptibility maps distinguish between different rock types, rather than distinguishing strength variability within those rock types. This is because the input data usually come from geological maps rather than geotechnical maps and measuring the rock strength at a high sample resolution, and depth, is both difficult and expensive. Consequently, topographic data are generally of much higher quality than strength data, despite the importance of both.

Material strength, structure and topography precondition all landslides and are thus important information for assessing hillslope failure susceptibility (i.e. the spatial distribution of potential failures). However, measured instantaneously, they do not provide a useful way to predict the timing of those failures. For failure to occur, the FoS needs to drop below unity. This may be caused by an increase in hillslope angle (considered above), but is most often due to either a decrease in the intrinsic strength or a gradual or sudden increase in an external stability factor. The rest of this chapter focuses on these time-dependent factors.

2.3.2 Strength degradation

The intrinsic strength of a hillslope can reduce slowly over time as a result of alteration (weathering) of the hillslope materials or strain accumulation.

Strength degradation can involve any combination of a range of weathering processes, including those of chemical decomposition of materials or stress–strain processes involving the growth of fractures or development of a failure surface. The gradual loss of strength, or strength degradation, is a phenomenon experienced by all slopes; therefore, much can potentially be gained from studying the processes involved. Despite its importance, strength degradation is perhaps the least understood of all landslide causes because it operates internally, making it difficult to observe and quantify. Strength degradation plays a fundamental role in controlling when a hillslope will fail. This role could be visualised in a model such as that of Fig. 2.3, where the gradient of the time-varying stability line would influence the time that $FoS = 1$ is reached. In the absence of other external factors capable of triggering failure, strength degradation may itself eventually trigger failure. Most of the time, its role is probably to reduce stability, increasing the likelihood for triggering by other instability factors.

Strength degradation, or its various processes (e.g. static fatigue, permafrost degradation and weathering), has often been cited as the possible cause of failures that had no apparent trigger. Indeed, there are numerous failures which have occurred without any apparent trigger (Wieczorek and Jäger, 1996), even when the events have been witnessed (e.g. Hauser, 2002; Lipovsky et al., 2008). This is not to say that strength degradation is the only possible trigger for many of these hillslope failures; there may be changes in the slope by more commonly observed processes that have, for whatever reason, gone undetected or unrecognised, such as unmonitored seepage pressures. For example, recent analysis of temperature conditions by Allen and Huggel (2013) provides support for a plausible alternative hypothesis for the trigger of the 1991 Mt Cook Rock Avalanche in New Zealand, which had previously been cited as an example of a hillslope failure triggered by strength degradation, or more specifically, a drop in the intrinsic strength of the hillslope (Glade and Crozier, 2005; McSaveney, 2002; Petley, 2011). Allen and Huggel argued that the trigger was pressurised meltwater. The meltwater was generated during 4 days of extremely warm temperatures, which was then entrapped and pressurised behind seepage outlets that were blocked during two subsequent days of freezing conditions. Whilst careful monitoring and scrutiny of processes may help to reveal the subtle triggers for landslides, strength degradation is no doubt an important process and likely to act as a trigger for some failures. Further, as Petley (2011) points out, it may also help explain why some landslides are triggered not in a major event but subsequently in a more minor event; i.e. initial damage may have been sustained or progressive failure initiated in the first event.

2.3.2.1 *Stress-induced fatigue*

Stress-induced fatigue encompasses all processes of stress-induced degradation in material strength that operate at stresses below the instantaneous ultimate failure strength of the material (e.g. Attewell and Farmer, 1973; Cruden, 1974; Lajtai and Schmidtke, 1986; Potyondy, 2007). It typically operates at the microscopic scale of inter-particle forces and involves the nucleation and growth of fractures within brittle materials. Local failure is thought to be possible at stress magnitudes below the ultimate failure strength because of concentration of stresses around structural imperfections such as micro-fractures; the growth of such fractures under these conditions is referred to as subcritical fracture propagation. Within a perfectly homogenous and uniform mass of material, the stresses may be distributed approximately equally within the material. However, all materials contain microscopic imperfections and many rock masses are heterogeneous and contain fractures and joints arising from previous exposure to high-magnitude stresses (e.g. tectonic stresses in excess of the ultimate failure strength). In addition, rocks may exhibit unbalanced stresses that develop through rock-forming processes (e.g. cooling or burial) or secondary processes (e.g. tectonic stresses). These imperfections and stress imbalances result in stress concentrations. These concentrations can cause micro-fractures to nucleate, which further increases the nonhomogeneity of the stress distribution. The stresses become concentrated at fracture tips, causing fracture propagation and further stress concentration. This positive feedback loop can, in theory, involve a nonlinear (rapidly accelerating) increase in stress concentrations as fractures propagate and result in the catastrophic failure of a material.

Stress release fracturing can also occur from unloading of overburden or confining materials (Nichols, 1980). This release of stress can induce stress re-orientations, potentially leading to stress concentrations sufficient to cause fracture propagation. It is a process familiar in the mining and tunnelling industries where high confining stresses exist and can be altered relatively quickly, sometimes leading to rock bursts. In natural slopes, the same process occurs, often operating over a longer time period owing to the larger scale of the material and the longer time frames by which natural processes (erosion) unload the material. The exception to this may be the unloading induced by other mass movements, which can trigger stress release behind the release surface. Stress release fracturing may be important in priming or triggering failures in slopes that have undergone glacial erosion (McColl, 2012).

Static fatigue is the gradual damage to a material as a result of the constant application of stress; for slopes, this stress arises from the self-weight stresses (acceleration of gravity acting on the mass) in addition to any constant tectonic

or internal/residual stresses. The steeper or higher a hillslope, the greater the self-weight stresses that are in operation. Thus, the steepening, loading or heightening of a slope will likely facilitate faster rates of sub-critical fracture growth and static fatigue of hillslope materials. Pore fluids (e.g. water) also exert stresses, through chemical reactions, at fracture tips and facilitate the sub-critical growth of fractures (this is referred to as stress corrosion). Experiments show that the introduction of water to mechanically stressed fractures can cause rapid increases in fracture propagation (e.g. Voigtländer et al., 2018), and so a combination of mechanical and environmental processes are likely to control static fatigue in hillslopes. Because the positive feedback mechanism of stress-fatigue causes an acceleration of fracture propagation, it may be possible for a hillslope to fail ‘out of the blue’ with no apparent change in the hillslope conditions or external factors (e.g. change in slope angle, rainfall or seismicity). Progressive failure mass movements in brittle materials are thought to involve a stress-fatigue process, involving a strain-controlled development of a through-going failure surface from the coalescence of fractures or breakage of intact rock bridges (Petley et al., 2005) (Fig. 2.5). This is a process that may operate without external stimuli, involving internal deformation of material, even whilst the global FoS remains above one.

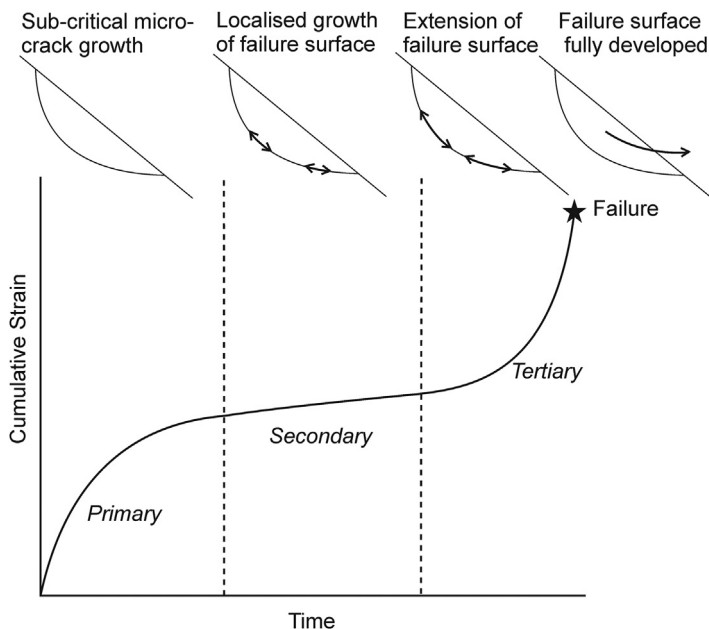


FIGURE 2.5 Schematic representation of the development of a through-going failure surface during the different stages of creep that govern progressive failure. Based on Petley et al. (2005) and Froude (2011).

Progressive failures therefore may occur without any other apparent trigger, but fortunately some warning may be possible if the pre-failure deformation can be detected. [Petley et al. \(2005\)](#) discuss how strain monitoring can be used to forecast the timing of progressive failures or failures that are undergoing accelerating creep; but, this relies on the strain being detected. Identifying potential failure sites before they show detectable strain or geomorphic signs of imminent hillslope failure (e.g. bulging or fractures at the ground surface) presents a greater challenge. This challenge may be met with advances in regional or even global-scale remote-sensing approaches such as satellite-borne InSAR or photogrammetry.

Cyclic-fatigue involves the fatigue of materials in a fashion similar to that of static fatigue; but, instead of being static, the stresses acting on a material are fluctuating (cyclic) stresses. Laboratory experiments show that the number of stress cycles a material can withstand is reduced with increasing cyclic stress amplitude, and there tends to be a stress limit below which no amount of cycles will induce fatigue. Because of the high amplitude stress cycles produced by earthquake shaking, much research on cyclic-fatigue in slopes has been studied with respect to earthquake ground motions (e.g. [Gischig et al., 2016](#); [Moore et al., 2012](#)). However, as with static stress fatigue, fractures may propagate at sub-critical stresses. Stress cycles in slopes operating at much lower frequency or magnitude than those induced by strong earthquakes could arise from a variety of other processes, such as vibrations from machinery, pore pressure fluctuations ([Eberhardt et al., 2016](#)), thermal changes (e.g. [Collins and Stock, 2016](#); [Gischig et al., 2011](#); [Grämiger et al., 2018](#)), snow or ice loading and unloading ([Grämiger et al., 2017](#)), changes in atmospheric pressure and wind gusts or tidal forces. Damage sustained from such processes, including earthquakes, may be difficult to detect. It is thought that repeated earthquake shaking that does not trigger failures immediately may nonetheless weaken slopes over time ([Gischig et al., 2016](#)) and contribute to failure during a subsequent (and possibly smaller) earthquake (e.g. [Parker et al., 2015](#)) or as a result of other triggers that would not have otherwise been sufficient to cause failure ([Davies and Moretti, 2021](#), [Hancox, 2010](#); [Moore et al., 2012](#)). However, the priming of hillslopes by earthquakes may not occur in all circumstances ([Brain et al., 2017](#)).

2.3.2.2 Chemical weathering

Chemical alteration of rock and sediments can reduce the shear strength and equilibrium angle of slopes ([Durgin, 1977](#)). As described earlier, the chemical processes of stress corrosion can facilitate fracture growth in brittle materials, often acting in combination with mechanically induced fatigue. A variety of chemical weathering processes can contribute to the reduction in material strength of brittle and nonbrittle materials and sediments; for example, dissolution, hydrolysis, hydration and oxidation. These processes may reduce

material strength by weakening the intact particles or the contacts between particles (especially along fractures or other water conduits) or change the sensitivity of a material to water (e.g. through the production of some types of clay minerals). Chemical weathering is most rapid in warm humid (i.e. tropical) environments, but it occurs in all environments.

The weathering of hillslope materials controls the stability of the slope and sets the frequency (rate) of regolith-stripping landslide activity in bedrock landscapes where the weathering rate of bedrock limits the production rate of regolith and soils. Weathering will also influence the type of landslide likely to occur as the structure of the hillslope material changes. [Durgin \(1977\)](#) relates the degree of weathering in granitic rocks to the dominant types of landslides. For example, rockfalls are most common on unweathered jointed rocks, whereas debris flows become more prevalent once weathering has begun to disintegrate the rock granules and completely weathered granites without any structural control host rotational failures (slumps). This dependence of hill-slope strength on weathering means that landslide susceptibility based on rock type alone may be misleading, especially because different rock types and rock masses weather at different rates. Indeed, [Durgin \(1977\)](#) describes how granite rocks in some regions are relatively unweathered and provide the most stable slopes (i.e. host the least landslides) of all rocks in that region, whereas in regions where chemical weathering is intense, granites can be the most prone to instabilities. The degree of weathering is thus an important factor to quantify. However, the relationship between chemical weathering and strength degradation is not necessarily straightforward. [Fan et al. \(1996\)](#) determined that the strength of fresh rocks with well-developed clay fabrics actually increased during initial weathering because of changes in fabric. Weathering-induced strength degradation did not occur until the sediments themselves were altered. Initially, cohesionless materials may increase in strength as weathering progresses because of the additional cohesion provided by weathering products.

2.3.2.3 Cold environment processes

Permafrost degradation: In environments cold enough to experience sustained freezing conditions (permafrost), another mechanism of strength degradation can exist, controlled by thawing of frozen rock or soil. Frozen materials tend to have higher shear strength than their unfrozen equivalents, either from stronger solid–solid contacts or from the strength of ice intercalation ([Krautblatter et al., 2013](#)). As earth and rock materials warm and thaw due to climate change, the drop in strength may be sufficient to shift a stable hillslope to a marginally stable state or trigger failure. Further, as ice melts, percolated meltwater can become pressurised and lower the effective normal stresses acting on potential failure surfaces or between particles.

A growing body of empirical evidence suggests a link between warming temperatures, or extremes in warm temperatures, and the historical activity of landslides (e.g. Allen and Huggel, 2013; Gruber and Haeberli, 2007; Lewkowicz and Way, 2019). For rock slopes, Krautblatter et al. (2013) have developed an ice–rock mechanical model to explain the loss in strength during thawing of permafrost; and notably, their model explains why permafrost degradation can trigger deep-seated rock slope failures as well as failures in the near surface. Interestingly, their model considers strength degradation arising from a combination of static fatigue, progressive failure and a drop in the intrinsic (frictional) strength of rock and ice–rock contacts.

Freeze-thaw processes: Weathering associated with repeated melting and re-freezing of interstitial water has long been recognised as another preparatory factor in jointed rock masses (Matsuoka and Murton, 2008). The volumetric expansion of water upon freezing (ice-wedging), coupled with the movement of water towards the freezing front (ice segregation), can generate sufficient pressures to induce fracture propagation (Matsuoka and Murton, 2008; Selby, 1993). Draebing and Krautblatter (2019) show that whilst volumetric expansion can generate super-critical fracture growth in alpine rock slopes under rare conditions, ice-induced fracture propagation mostly occurs at sub-critical stresses from both volumetric expansion and ice segregation processes.

Several researchers have suggested a link between freeze-thaw processes and modern rates of rockfall activity (Gruber et al., 2004; Matsuoka, 2008; Matsuoka and Sakai, 1999; Noetzli et al., 2003; Ravanel et al., 2010) and possibly even a link to large-scale rock slope failures (Davies et al., 2001; Wegmann and Gudmundsson, 1999). In addition, the likely intensification of freeze-thaw conditions, possibly in combination with permafrost degradation, during the last deglaciation may have led to more enhanced rockfall activity and talus development during this period (e.g. Rapp, 1960).

2.3.2.4 Discussion

It is likely that for all hillslopes strength degradation plays a role in reducing the critical failure threshold required for a host of other more recognisable triggers, or may itself be the final failure trigger. To that end, if strength degradation could be quantified accurately, along with the driving and resisting stresses operating for every hillslope, accurate landslide predictions would become much more achievable. The reality is that we are far from achieving a robust understanding of strength degradation or a means to accurately predict its effect. Two of the limitations to understanding strength degradation are the difficulty of observing the processes involved and the impracticality of knowing the entire history of the hillslope which led to the slope strength and stress conditions at a given point in time.

A fuller understanding and predictive capability may best be achieved by combining multi-disciplinary approaches. For example, undertaking detailed engineering geological mapping and strength testing (e.g. [Brideau et al., 2009](#)); using powerful numerical techniques to model progressive failure (e.g. modelling progressive failure; [Eberhardt et al., 2004](#); [Locat et al., 2013](#)) or stress corrosion/static fatigue processes (e.g. [Potyondy, 2007](#)); using geophysical approaches, such as measuring the micro-seismicity generated by precursory fracture propagation (e.g. [Amitrano et al., 2010](#); [Spillmann et al., 2007](#)) or repeat imaging of the material properties to assess reductions in strength through time (e.g. spectral analysis of earthquake ground motion; [Moore et al., 2012](#)). Synoptic models that attempt to incorporate a range of processes, such as that of [Krautblatter et al. \(2013\)](#) for permafrost degradation, hold promise for holistic strength degradation models in other types of environments. Because of its pervasive applicability, developing better predictive capability for strength degradation is arguably one of the greatest and potentially most fruitful challenges for hillslope stability research.

2.3.3 Groundwater changes

Dynamic triggers are factors that are transient and bring a hillslope from a marginally stable state to failure or movement when they temporarily exceed the failure threshold. Changes in groundwater (and seismic ground accelerations discussed in the next section) are the most commonly recognised dynamic triggers. A useful point to keep in mind is that most slopes experience numerous dynamic events without undergoing failure, and it is not necessarily the largest of these that causes final failure. The effectiveness of those dynamic events in triggering failure depends not just on event magnitude but on the stability threshold, which can reduce through time as a result of other preparatory factors (e.g. strength degradation).

Groundwater fluctuations arise from either a change in the infiltration of water into (e.g. from rainfall or snowmelt) or evapotranspiration out of a hillslope, from a volumetric change in porosity or from a change in the drainage conditions of the slope (e.g. changes in seepage pressures or flooding at the base). Groundwater fluctuations can alter stability in the following ways:

1. Groundwater fluctuations drive changes in porewater pressures, which change the effective normal stresses and therefore the shear strength of potential failure surfaces or particle contacts.
2. The generation of phreatic surface slopes or groundwater pressure gradients which induce flow or seepage within a hillslope. Flow or seepage can add a driving force (e.g. drawdown following rapid lowering of lakes or reservoirs at the base of slopes).
3. Saturation with water can change the weight and inherent strength of materials (e.g. saturation of soils and swelling clays).

The relationship between groundwater levels (or more often the magnitude or intensity of rainfall) and landslide activity has been investigated empirically, with the establishment of triggering thresholds, and deterministically using slope stability models.

Rainstorms are a major trigger of landslides (in particular shallow ones within weathered material), often generating multiple failures in one event (Crozier, 2017; Van Asch et al., 1999). Whilst good empirical relationships can be established between rainfall and landslide triggering thresholds (Glade, 1998) – especially when antecedent soil moisture can be taken into account (Crozier, 1999) – the processes contributing to scatter in these relationships are difficult to identify. Uncertainty with rainfall thresholds arises largely from uncertainty in the infiltration, flow pathways and drainage conditions in the hillslope and the specific response of hillslope materials. For example, where effective cohesion, rather than only the frictional properties of a soil, strongly influences stability, it may be peak moisture content rather than peak pore-water pressures that trigger failure (Hawke and McConchie, 2011). Surface topography, the topography of the failure surface (usually the bedrock–soil interface) and spatial variability in the permeability of the hillslope materials and their substrate have a strong influence on the spatial relationship and timing between rainfall and groundwater responses. For example, the confluence of water along the bedrock–soil interface of bedrock hollows or fossil gully features (colluvium-filled bedrock depressions) compounds the saturation of materials in these hollows, and consequently these are common hillslope failure sites (Crozier et al., 1990; Wilson and Dietrich, 1987). Whilst identifying and modelling the surface topography of a hillslope is relatively easy, the topography of the bedrock–soil interface is more difficult to identify. The bedrock–soil interface is not always impermeable as often assumed. The underlying bedrock can store and release water into the soil during rainfall events, and thus may have a considerable influence on both the timing and distribution of soil saturation during a rainfall event (Brönnimann et al., 2013).

Despite uncertainties in the scatter found in relationships between rainfall and landslide activity, recent analysis of empirical studies has demonstrated that in general the total rainfall generated in a storm is a very strong predictor of landsliding (Marc et al., 2018). Further, for rainfall-triggered landslides, the most recognised factor modifying the relationship between soil saturation and rainfall is vegetation. Although preferential flow along living or decayed tree roots can change soil saturation through improved drainage (Ghestem et al., 2011), vegetation removal, which removes the interception of rainfall and evapotranspiration, is perhaps the most common preparatory factor for the development of shallow landslides and has been recognised as a major contributor to human-induced global soil erosion (Glade, 2003).

Whilst rainfall is more commonly attributed to triggering shallow landslides during storm events, rainfall can cause the failure or movement of deep-seated landslides, but usually through longer-duration or seasonal rainfall

patterns rather than individual rainfall events. This is due to less permeable and less responsive bedrock hydrology, compared to the thin surficial soils of shallow landslides. For large, slow-moving landslides, porewater pressure and movement are generally positively correlated. However, the advent of high-resolution surface and subsurface monitoring of movement and groundwater fluctuations has shown that this is not always the case. Other internal (e.g. strain hardening during shearing or development of negative porewater pressures in zones of extension), geometrical/topographical or nonfrictional (e.g. rheological) mechanics can modify or even reverse this relationship (Corominas et al., 2005; Massey et al., 2013, 2016; van Asch et al., 2009).

Understanding how water can destabilise hillslopes is important in the context of anticipated global climate changes (Jakob, 2021). Whilst it is clear that climate change can modify rainfall patterns, evaporation and vegetation, and that these factors can influence landslide activity, the realisation of a link between climate patterns and landslide activity is difficult to assess (Crozier, 2010). However, over pre-historical (i.e. glacial–inter-glacial) timescales, landslide activity appears to correspond to periods (e.g. inter-glacial climate optima) of heightened humidity (Pánek, 2019), and future projections suggest mostly heightened (but some reduced) rainfall-induced landslide activity in certain parts of the world (Gariano and Guzzetti, 2016; Jakob, 2021).

2.3.4 Ground shaking

Seismic ground shaking or artificial sources of ground vibrations are another type of dynamic failure trigger. Earthquake shaking, as opposed to artificial sources, is the focus of this section because earthquakes are more capable of releasing seismic energy strong enough to cause hillslope failure (Fig. 2.6). There are three main (and nonexclusive) ways that a slope failure can arise

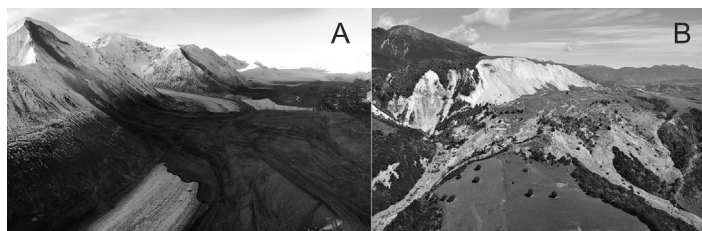


FIGURE 2.6 (A) Examples of coseismic slope failures triggered by the 2002 Denali earthquake, Alaska. These rock avalanches failed from the top of the slope, as is typical of earthquake-triggered landslides likely due to topographic amplification, and travelled on to the Black Rapids Glacier. (B) Example of slope failure triggered by the 2016 Kaikoura earthquake, New Zealand. The Papatea fault, which ruptured in the earthquake, cuts through the back of the source area; as with many of the landslides triggered by this earthquake, proximity to a fault seems to be major control of landslide distribution (Massey et al., 2018). (A) Photograph: Rod March, US Geological Survey. (B) Photo: Dougal Townsend, GNS Science/EQC.

from earthquake shaking, all dependent on the duration and/or the intensity of shaking:

1. Sudden ground accelerations that induce instantaneous shear stresses in excess of the mobilised shear strength.
2. Cyclic fatigue-induced propagation of fractures (at sub-critical stresses), leading or contributing to the formation of a failure surface.
3. Cyclic shaking causing an increase in porewater pressures.

Despite general uncertainty in the specific processes governing earthquake-triggered failure, good relationships between earthquake magnitude or intensity and the density or distribution of landslides have been established from a number of earthquakes (Dai et al., 2011; Hancox et al., 2002; Keefer, 1984; Meunier et al., 2007). In general, the number and spatial extent of landslides scale with earthquake magnitude, but are also dependent on the static factors influencing failure susceptibility (e.g. lithology and slope angle; Dai et al., 2011). Earthquakes appear to be responsible for triggering many of the larger known landslides on Earth, and stronger ground motions or locations closer to source faults tend to be associated with larger coseismic landslides (Valagussa et al., 2019).

The specific spatial distribution, types of landslides and their size seem to depend on a number of local seismic site effects and conditions. For example, for landslides triggered by the November 2016 M_w 7.8 Kaikoura earthquake in New Zealand, the distance to fault was a more powerful predictor variable than modelled shaking intensity, potentially due to the interaction of higher-frequency ground motions and topographic amplification nearer to the faults (Massey et al., 2018) (Fig. 2.6). Seismic ground motions depend on a variety of seismic wave characteristics interacting with the unique geologic, structural and topographic conditions at any given site (Aki, 1988; Alfaro et al., 2012; Buech et al., 2010; Geli et al., 1988; McColl et al., 2012; Meunier et al., 2008). Weakened and fragmented material associated with existing or incipient landslides (i.e. zones of increased shear wave velocity) may even modify and amplify the shaking experienced on those parts of the hillslope (Burjánek et al., 2012; Moore et al., 2011). The hydrological conditions at the time will also influence the distribution of coseismic failures; a high groundwater table may lower the static FoS but also make any seismically induced rise in groundwater more effective.

2.4 Summary and conclusion

The geological and geomorphological controls on hillslope failure (e.g. the type, strength, and structure of material, and the slope geometry) play an important role in predisposing hillslopes to failure and controlling the likely spatial distribution of failures. These factors also largely determine which type (style) of failures will occur and have a strong bearing on the threshold for

stability on which other stability factors operate. Spatial and geostatistical analyses using GIS approaches help to elucidate the strength of the relationships between various stability factors and landslide failure and are a useful tool for calculating landslide susceptibility. However, even though it is possible to identify the slopes most likely to fail in a regional study, predicting the frequency of hillslope failures on a regional scale is a considerably harder challenge. Many of the slopes investigated in a regional-scale assessment of susceptibility may share apparently similar predisposing factors; but precisely which slopes will fail during any given period of time or with the application of an external trigger may vary beyond our capability of prediction without resorting to methods of statistical probability. Even local-scale investigations of single slopes may fail to predict failure, due to the difficulty in measuring the full complexity of natural slope systems and perhaps especially the processes of strength degradation. This challenge will continue to be met through advances in knowledge, such as better recognition and investigation of the multitude of strength degradation processes, better methods becoming available and applied for observing, monitoring and modelling slopes and further attempts made to develop long-term time-dependent models of instability. These advances would potentially allow the quantification and useful application of stability models as proposed schematically in [Fig. 2.3](#).

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